

2025-10-22 MXCuBE AutomationWG

Agenda:

1. Definition of the output parameters from X-ray centering
2. The plan of the upcoming tasks

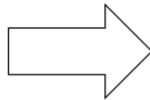
Notes:

As an introduction the following slide was presented to open the discussion on the output. We already agreed on the input parameter as stated in the figure. Thanks to Rasmus and Martin, MXLIMS now has a model for X-ray centering and especially for the volume to scan. (see https://github.com/rhfogh/mxlims_data_model/blob/main/ReleaseNotes.md)

X-ray centering

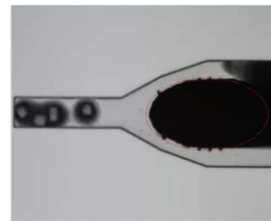
User input parameters:

1. Sample parameters:
 - Volume to scan as implemented in MXLIMS
2. Beamline parameters:
 - Standard as the one from the DC
 - Default values in the configuration of the BL
3. Acquisition parameters:
 - Standard as the one from the DC
 - Default values in the configuration of the BL



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Output parameters



Point 1

- From the volume to scan in the sample parameters, the X-ray centering deconvolutes 2D scans with different angle orientations.
- From these 2D-scans, MXLIMS abstracts a 3D volume and sub-volumes with XYZ coordinates.
- MXLIMS is implemented with the volume and sub-volume outputs as a list of point cloud to define all the samples that are on the sample support.
- Each point cloud (X, Y, Z) is model with a set of quality indicator.
- The set quality indicators needs to be defined to take advantage o. all the capabilities available in the scoring program that would be used to analyzed the images produced by the 2D-scans

- The set of quality indicators should reflect the homogeneity of a single crystal and or the presence of multiple crystal on the same support.
- The minimum set of quality indicator includes:
 - sum of intensities
 - number of spots
 - resolution
 - lattice meaning that the lattice at the position XYZ is equivalent to the lattice at position X'Y'Z'
- The set of quality indicator can be later on expanded with new software becoming available.
- It would be better the model in MXLIMS is fully determined beforehand with all use cases instead of having a dictionary (XML or JSON) that will be filled up differently by each sites.
- Processing of the images is done on a cluster that is not the machine running MXCuBE and the processing is launch on the fly and interdependently. The X-ray centering should be able to launch different function with the mesh as an input and will be able to come back with position, centroid, etc as the site would prefer. This processing needs to be part of MX-CuBE and the X-ray centering procedure
- Rasmus offers to put up a survey to find out what is the minimum set of quality indicators in all sites and the different sites will have to provide what kind of quality indicator they use to do their X-ray centering. Also if they use this as a tool for multi-crystal centering

Point 2

- Abstract diffractometer is under development at ESRF. It should be up and running by beginning of 2026
- X-ray centering will be based on Abstract diffractometer
- Processing will have to be further discussed for implementation until May 2026